Rozklad na součin - vzorce

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

a) $x^3 - 27 = x^3 - 3^3 = (x-3)(x^2 + 3x + 9)$

b) $125 + a^3 = (5 + a)(25 - 5a + a^2)$

c) $64x^3 + 1 = (4x + 1)(16x^2 - 4x + 1)$

d) $27a^3 - 216 = (3a)^3 - 6^3 = (3a - 6)(9a^2 + 18a + 36)$

e) $125a^3 - 8b^3 = (5a - 2b)(25a^2 + 10ab + 4b^2)$

f) $125 + a^6 = (5 + a^2)(25 - 5a^2 + a^4)$

g) $(3x - 2)^3 + 27 = (3x - 2 + 3)[(3x-2)^2 - 3(3x-2) + 9] = (3x+1)(9x^2 - 21x + 19)$

$A^3 + B^3$; $9x^2 - 12x + 4 - 9x + 6 + 9$

h) $(4a + 3)^3 - 8 = (4a + 3 - 2)[(4a+3)^2 + 2(4a+3) + 4] = (4a+1)(16a^2 + 32a + 19)$

$16a^2 + 24a + 9 + 8a + 6 + 4$

i) $(2x + 3)^3 - x^3 = (2x + 3 - x)(4x^2 + 12x + 9 + 2x^2 + 3x + x^2) = (x+3)(7x^2 + 15x + 9)$

j) $(x - 3)^3 - (x + 2)^3 = (x - 3 - x - 2)[(x-3)^2 + (x-3)(x+2) + (x+2)^2] =$

$A^3 - B^3$; $x^2 - 3x - 6 + 2x$

$= -5(x^2 - 6x + 9 + x^2 - x - 6 + x^2 + x + 4) =$

$= -5(3x^2 - 3x + 7)$

$k)\ (2a-1)^3+(a+3)^3 =$ $(2a-1+a+3)\left[(2a-1)^2-(2a-1)(a+3)+(a+3)^2\right] =$

$= (3a+2)(4a^2-4a+1-2a^2-6a+a+3+a^2+6a+9) =$

$= (3a+2)(3a^2-3a+13)$

$l)\ (3x-y)^3-(x+2y)^3$

$i)\ 45x^2ya^2-5x^2y$

$j)\ x^2(x+1)-(x+1)$

$k)\ 4x^2(x-1)+(1-x)$

$l)\ x^3(x-3)+x(3-x)$

- $m)\ a^4(a-1)+4a^2(1+a) =$ $a^4(a-1)-4a^2(a-1) = (a-1)(a^4-4a^2) =$

$= (a-1)a^2(a^2-4) = a^2(a-1)(a-2)(a+2)$

- $p)\ 27x^4-x =$ $x(27x^3-1) = x(3x-1)(9x^2+3x+1)$

$x^4-y^4 = (x^2-y^2)(x^2+y^2) = (x-y)(x+y)(x^2+y^2)$